PalmSource™ 99

Symbol Technologies, Inc.
SPT 1500 and SPT 1700 SDK

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Symbol Palm™ Terminal SDK

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- New Models of the SPT
  - 1500
  - 1700
  - 1740
- Contents of the SDK
- New Additions to the SDK
  - Printing Capability
  - Magnetic Stripe Reader
  - Spectrum24 Wireless Radio

SPT SDK

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Palm OS® versions

- SPT 1500 uses Palm OS, version 3.02
- SPT 1700 uses Palm OS, version 3.2

SPT SDK

- Scanning
- Printing
- Magnetic Stripe Reader
- Spectrum24 Radio

SPT SDK Contents

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- SDK is the same for the SPT 1500 and SPT 17xx
- What is the same?
  - API has not changed
  - Scan Manager Library
  - Flash Utilities
  - Development Tools
    - CodeWarrior
    - Satellite Forms
    - Etc.

SPT SDK—Enhancements

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- Additions to the SDK
  - Printer API
  - Magnetic Stripe Reader API
  - S24 Wireless Radio API
  - Flash Utility for SPT 1500 and SPT 17xx family
  - Sample Code for Scanning, Printing, MSR and S24
- Documentation
- Libraries
  - Printing
  - MSR
  - S24

Adding Files to a SPT

- Flash into ROM
  - By using the FlashUtilities:
    - rombuild.exe
    - flnstl.exe
- Synchronization
  - HotSync
  - Using third-party tools
- Wireless Synchronization

Additions to the SPT SDK

- Printing
- Magnetic Stripe Reader
- Spectrum24 Radio



Why Printing?

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- Add printing capability for barcode printers
- Add printing for receipt printers
- Allow users with commercial printers to print using Postscript and PCL

Printers Currently Supported

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- Label and Barcode Printers
  - Comtec
  - Eltron
  - Monarch
  - O'Neil
- Commercial Printers
  - PCL
  - Postscript

Printer API Features

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- High-level text and Graphics
- Generic write function
- Supports many printer types
- Ability to handle multiple transports
  - Serial
  - Ir

Architectural Overview

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- Shared library
- Printer capability file (printcap)

Ways to Change Printer Settings

- Add an entry to the PrintCap file
- Set values in the PTConnectSettings structure
- Send it directly to the printer with the ptWritePrinter command, for an “Unknown” printer

API Function Calls



- General Purpose Interface Functions
- Low-Level API Calls
- High-Level API Calls

General Purpose Interface Functions

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- Open
- Close
- Connect
- Disconnect
- Initialize
- Reset
- Query

- ptOpenPrinter
- ptClosePrinter
- ptConnectPrinter
- ptDisconnectPrinter
- ptInitPrinter
- ptResetPrinter
- ptQueryPrinter

High-level API Calls

Simple graphics and text

- Draw line
- Print buffer contents
- Draw rectangle
- Buffer control
- Control fonts
- Set initial buffer size
- Add text to buffer

- ptLineToBuffer
- ptPrintPrintBuffer
- ptRectToBuffer
- ptResetPrintBuffer
- ptSetFont
- ptStartPrintBuffer
- ptTextToBuffer

Lower-level API Calls

- Query PrintCap
- Write to Printer

- ptQueryPrintCap
- ptwritePrinter

  - Retrieve PrintCap Entries
  - Send user-formatted data to the printer

Data Structures

PTConnectSettings

represents serial or IrDA setting information

Defines:

baud rate

timeout, in system ticks

transport flags, as defined in SerialMgr.h

receive buffer size

Available Ir Baud rates: 9600, 57600, 115200

compliant with the Palm OS

Print Sample
Sample Code

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```
PTConnectSettingsPtr     pcSettings;


ptOpenPrinter();

ptConnectPrinter( NULL );

ptWritePrinter();

ptDisconnectPrinter();

ptClosePrinter
```

SPT SDK

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MSR 3000 Features

- Attachment for the Symbol SPT-1700/1740
- Interfaces via the HotSync® port
- Powered from SPT-1700/1740
- Barrel connector for Charging SPT-1700/1740 Battery
- Reads One, Two , or Three Track cards



MSR 3000 Features

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- Decodes most common formats
  - American National Standard magnetic stripe encoding format (ISO-7811)
  - AAMVA recommended format (Driver's Licenses)
  - California DMV magnetic stripe encoding format
- Generic and Raw Decode modes for non-standard formats

Software Components

- MSR 3000 Firmware
  - Performs MSR decode
  - Many Configurable options
  - Optionally parses data
- SPT-1700 MsrMgrLib shared library
  - API to configure and control MSR 3000
- Configurator Program
  - Offline tool to generate MSR 3000 configuration

Firmware Features

- **Buffer Mode**—Indicates whether MSR3000 will buffer data or send immediately
- **Terminator**—Character following last track of data, to separate card reads; CR/LF, CR, LF or None
- **Preamble**—String to be added prior to first track of data
- **Postamble**—String to be added following last track of data
- **Track Selection**—Selects which tracks the MSR3000 will decode

Firmware Features (Cont.)

- **Track Separator**—Character used to separate returned tracks
- **LRC**—Indicates if a LRC check character should be sent by the MSR3000
- **Data Edit**—Enable/Disable Data editing capabilities of MSR3000

Firmware Features (Cont.)

- Firmware settings stored in Flash of MSR 3000
- Since the MSR 3000 can be removed from the SPT-1740, you cannot assume the setting are as desired

MSR 3000 API

- The MSR Manager shared library API allows SPT 1740 application programs to control and receive data from the MSR 3000 Magnetic Stripe Reader
- A typical application program uses the MSR Manager shared library to do the following:
  - Open the MSR
  - Configure the MSR
  - Handle MSR data or error messages received from the MSR 3000 MSR
  - Close the MSR
- Implemented by MsrMgrLib.prc shared library

MSR Demo Program

- Demo application program included with the SDK
  - Executable, Source, and Project files
- Uses all of the API functions and demonstrates:
  - Using the API to set and get MSR 3000 parameters
  - Handling MSR data
  - Handling errors
- Uses the SPT 1740 graphic interface to display and change all MSR 3000 settings

MSR Sample Program

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- Very simple Demo application program included with the SDK
  - Includes Executable, Source, and Project files
- Illustrates the minimum needed to add Magnetic Stripe Reading to an application
- Good Starting point for applications
- Walked through in MSR 3000 System Software Manual (Chapter 5)

Simple Application Flow

- *Call MsrOpen()*
- Prompt user to swipe a card
- When "*msrDataReadyKey*" keyDown event is received
  - Call MsrReadMSRUnbuffer()
  - Process Data
- *Call MsrClose()*

API Functions

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```
MsrOpen                    MsrSetTrackSeparator
MsrClose                   MsrSetLRC
MsrGetSetting              MsrSetDataEdit
MsrSendSetting             MsrSetAddedField
MsrGetVersion              MsrSetDataEditSend
MsrGetStatus               MsrSetFlexibleField
MsrSelfDiagnose            MsrGetDataBuffer
MsrSetBufferMode           MsrReadMSRBuffer
MsrArmtoRead               MsrReadMSRUnbuffer
MsrSetTerminator           MsrSetDefault
MsrSetPreamble             MsrSetDecoderMode
MsrSetPostamble            MsrSetTrackFormat
MsrSetTrackSelection       MsrSetReservedChar
```

Data Editing

- Allows the MSR 3000 to edit the data that has been read from a magnetically encoded card before data is transmitted to the application software
- The application software receives the information in the exact format expected
- Can be defined via API calls or through offline tool (Configurator)
- Data Editing disabled by default

Data Editing Features

- Rearrange the Data fields, within a track
- Insert Character Strings into the Output Data Record
- Fields, within a track, can be transmitted to the application software as many times as desired in any order
- Fields, within a track, can be selected for output or not
- Output Method [Matched or Unmatched]
  - In Matched, MSR 3000 only sends data that matches formula

Data Editing with Configurator

- Windows GUI utility to assist in controlling and configuring the MSR 3000 features
- Creates an include file with preset data structures
- Call *MsrSendSet()* function to send settings to MSR 3000
- Common Field Formats pre-defined
  - Credit Card Format
  - Driver's License Format Recommended by AAMVA
  - California Driver's License Format

Special Magnetic Card Format Support

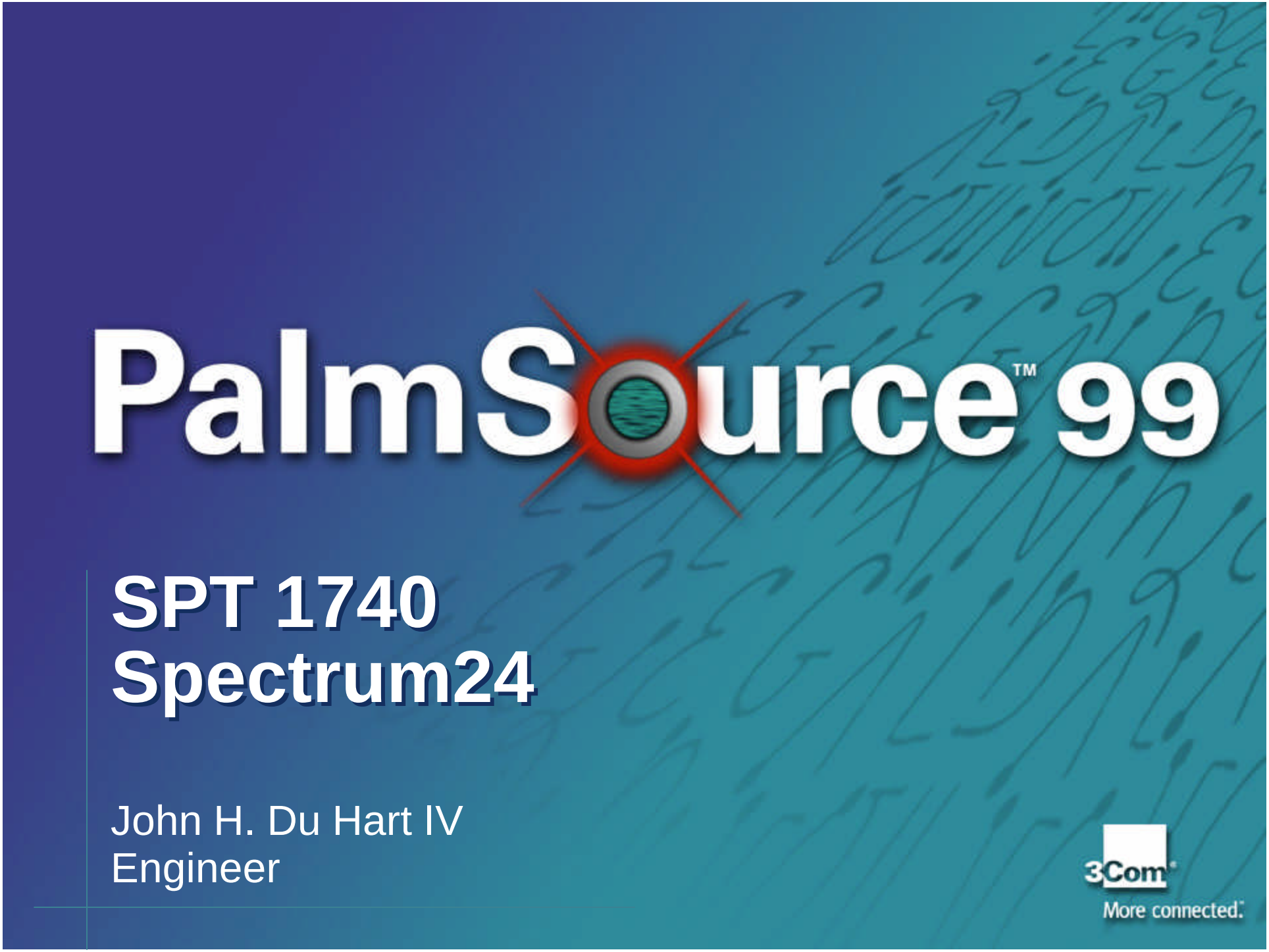
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- Provides Two unique features to support special magnetic card formats (Generic and Raw Decoders)
- Generic Decoder provides method for defining and reading non-standard 5 or 7 bit data formats
- Raw data decoder sends raw hex data, ASCII encoded, so the application can perform proprietary decode

SPT SDK

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- Spectrum24 Radio



SPT 1740
Spectrum24

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SPT 1740 Spectrum 24

- 1740 contains all the standard features of a Palm device
  - IRDA port maturity
  - Serial port
  - Standard keypad interface
  - Palm V type display
- Uses a Li-Ion rechargeable battery with a minimum 10 hours of battery life
- Integrated scanner
- Ruggedized

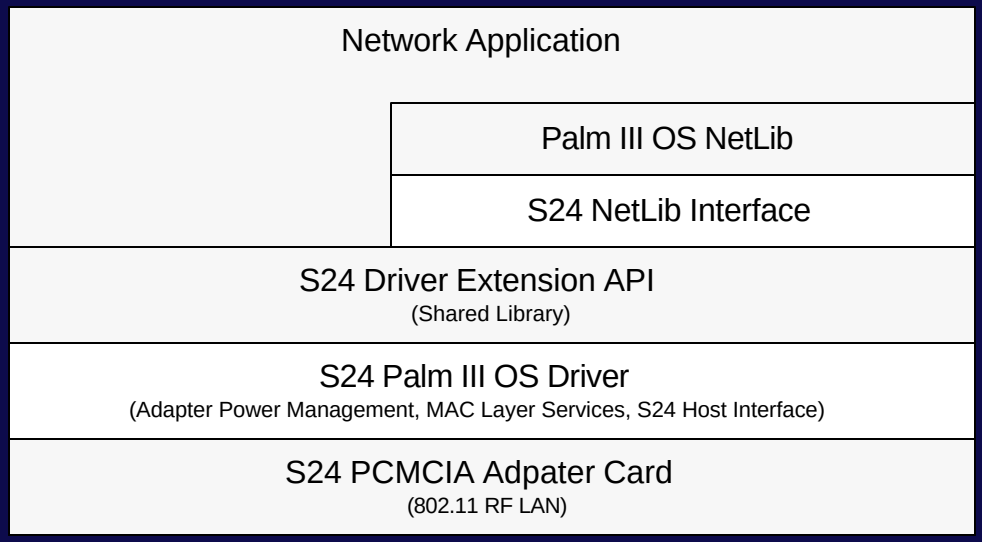
SPT 1740 Spectrum 24

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- Radio is a 1-Mbit spread spectrum device
- The radio and driver are 802.11 compliant
- Driver supports DHCP and ARP
- Receipt of UDP broadcast messages is supported
- Multicast messages are not supported

SPT 1740 Spectrum 24 Driver

- Fully integrated NetIF driver for S24 Radio



- All applications written for Netlib will function with Spectrum 24
  - Wireless Hot-Sync
  - Scout-Sync
  - AvantGo

SPT 1740 Spectrum 24 Setup

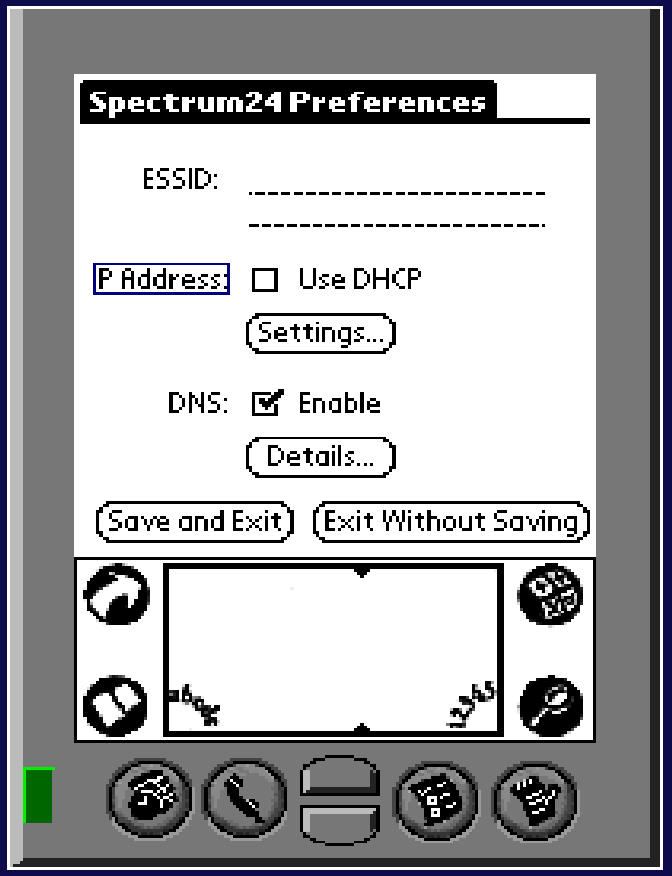
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- Specturm 24 is selected from the Network Preference panel service pick-list
- Setup is controlled by pressing the‘Details’ button on this screen
  - DHCP selection
  - DNS Servers
  - IP address
  - Sub-net Mask
  - Router address

SPT 1740 Spectrum 24 Setup

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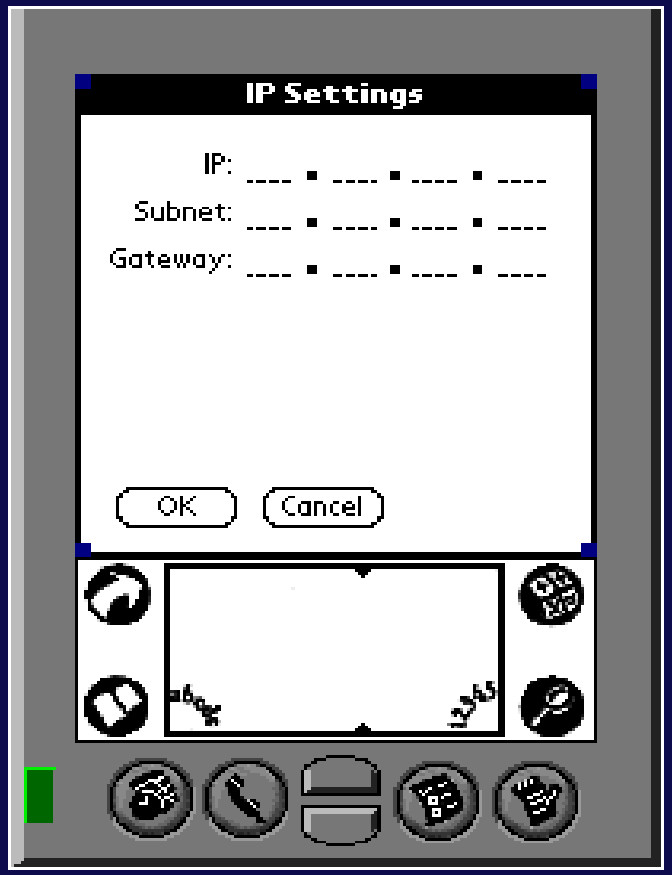
- From the S24 Pref screen you can set
  - ESSID for S24 network
  - IP/DHCP selection
  - DNS enable
- If DHCP is enabled IP, sub-net mask, router, and DNS information is gotten from the server



SPT 1740 Spectrum 24 Setup

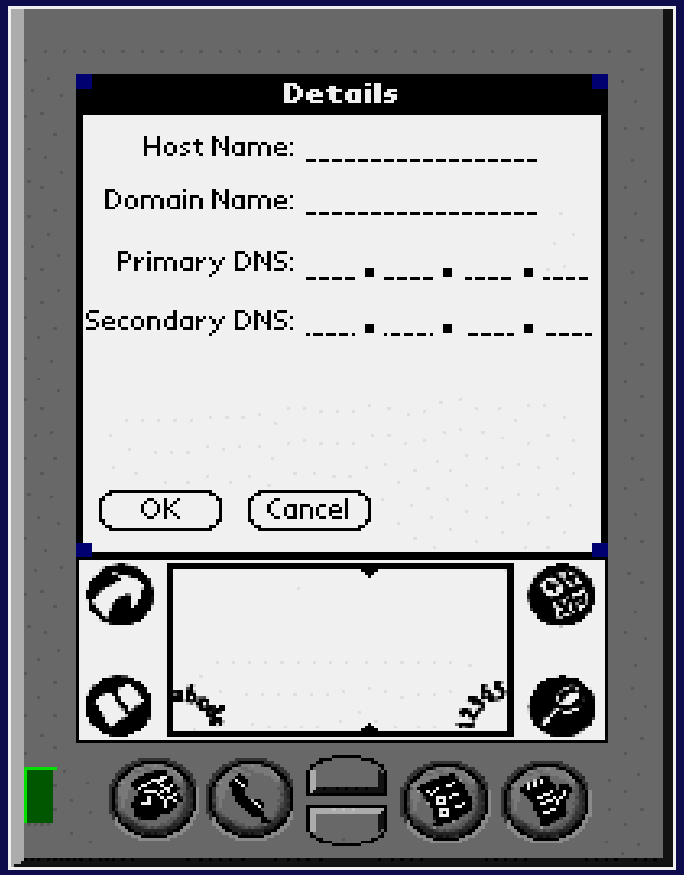
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- When DHCP is disabled you can setup your IP, subnet, and gateway by pressing the settings button



SPT 1740 Spectrum 24 Setup

- If DHCP is disabled you will need to setup DNS information if required by your application
- You can specify by name or IP address



SPT 1740 Spectrum 24

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- The Spectrum 24 driver includes a Extension API for radio specific functions
  - ESSID
  - Network association
  - Driver specific Preferences
  - Statistics
- Applications are not required to use any of the S24 Extension functions to work properly with the radio

SPT 1740 Power Management

- When a Low Battery is detected by the system the radio is shut down, and a S24BatteryError event is generated
- The application should inform the user that this event has occurred, and the net connection should be closed
- Attempting to reopen the connection before the battery has been recharged will cause this sequence to repeat itself

SPT 1740 Power Management

- Radio remains on when the unit auto-offs
  - When the unit auto-offs the radio is left on for a period of time, controllable by the application (default is 5 min.) This is know as napping mode. The unit will ‘wake-up’ on receipt of a message
- When a user powers off the unit with the power key it will cause the radio to be powered off immediately

SPT 1740 Spectrum 24 SDK

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- Symbol SDK includes documentation for the S24 Extensions API
  - Applications are not required to use any of these ‘Side Door’ functions to work properly with the radio
- Sample applications and source code are included
  - Calif. DMV app
  - Echo app


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